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A study of experimental privacy budget quantification in quantum differential privacy simulations

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ABSTRACT 150-250 palabras

INDEX TERMS Differential privacy, Quantum computing, Quantum noise

I. INTRODUCTION

ESTA va a ser la introduccion a la tesis. En esta sección pienso hablar de la motivación, la importancia de la privacidad diferencial, basicamente muy similar al documento de propuesta

II. BACKGROUND

En esta seccion se va a introducir todos los conceptos, ecuaciones y conocimiento necesario para entender el paper. Siguiendo el documento de propuesta propuesta vamos a tener:

A. DIFFERENTIAL PRIVACY

Las definiciones clasicas de privacidad diferencial

B. QUANTUM INFORMATION AND COMPUTATION

Las definiciones necesarias para entender el paper desde informacion cuantica: (estado, matriz de densidad, ruido cuantico, etc)

C. QUANTUM DIFFERENTIAL PRIVACY

Las definiciones de la privacidad diferencial cuantica y las de los algoritmos del paper de Zing para encontrar epsilon (dPD, PD)

III. RELATED WORK

El estado del arte muy similar a la propuesta donde se establece cual es el entorno de investigacion de la privacidad diferencial cuantica. En especial sobre la privacidad diferencial usando el ruido de decoherencia que es la forma que la estamos aplicando en el paper.

IV. DATA PROCESSING AND ANALYSIS

This section details the data processing and analysis methodology that was used in order to prepare the classical Statlog (German Credit Data) dataset [1] [2] for use in the quantum setting. Firstly we outline the dataset characteristics, variable selection procedures and lastly quantum encoding procedures for the selected data.

A. DATASET DESCRIPTION

The selected dataset for this investigation is the Statlog (German Credit Data) dataset created by Professor Hans Hofmann [1]. This multivariate dataset aims to classify people as a good or bad credit risk. The orginal dataset is composed of 1000 instances that span 21 variables (7 numerical, 13 categorical, and the target).

In the original dataset created by Professor Hofmann, all non numerical variables use labels to define their characteristics (e.g The status of a person whose checking account is less than 0 Deutsche Marks (DM) is A11) for that reason it was decided to use a curated version of the dataset [2] which transforms all the labels to their definition as a means of simplifying the readers understanding of the dataset by eliminating that intermediate layer of label and category.

Neither the original dataset nor the curated version have any problem in terms of completeness as all rows include all 21 variables nor uniqueness as there is no instance of an exact duplicate. For these reasons no special cleaning procedure was done.

B. VARIABLE SELECTION WITH PHIK

As the primary objective of this investigation was not to make a clasiffier but rather explore the capabilities that quantum computing could offer in terms protecting the input data, coupled with the fact that the simulation would at a maximum store 1000 $2^n \times 2^n$ density matrices (with n being the number of variables) and thus a larger n would likely need higher computational capabilities it was decided to select a subset of variables that in a true classifier setting would likely have the most incidence on training.

As mentioned previously however, most of the variables in this dataset are not numeric and thus a popular correlation such as the Pearson correlation analysis would be fruitless. For this reason it was decided to use a PhiK correlation coeficient [3] which is a stronger candidate to measure the association between a pair of variables that are not necessarily numerical.

The PhiK correlation test gives for a pair of variables their ϕ_K which is how strong the association is and a significance Z which states how statitically relevant that association really is. We used Baak et al. PhiK library to make this analysis between the 20 features of the dataset agains the objective variable and these were the results:

TABLE 1. ϕ_K correlation of each feature with the target and its statistical significance.

Feature	ϕ_K	Significance (z)	Significant
status_account_ord	0.512	10.951	Yes
month_duration	0.285	5.648	Yes
credit_amount	0.241	4.324	Yes
collateral	0.217	4.015	Yes
purpose	0.204	3.713	Yes
credit_history	0.197	6.900	Yes
status_savings_ord	0.147	5.297	Yes
age	0.142	2.172	No
status_and_sex	0.123	1.945	No
is_foreign_worker	0.109	2.529	No
years_employment_ord	0.098	3.047	Yes
housing	0.077	3.587	Yes
payment_to_income_ratio	0.075	1.045	No
other_installment_plans	0.063	2.901	No
secondary_obligor	0.041	1.843	No
telephone	0.021	0.663	No
n_credits	0.000	0.051	No
job	0.000	−0.291	No
n_guarantors	0.000	$-\infty$	No
residence_since	0.000	−1.090	No

C. ENCODING OF CLASSICAL VARIABLES INTO QUANTUM

La explicación de la metodologia de codificación de variables categoricas y númericas en qubits y los tradeoff que este proceso conlleva.

V. NOISE PARAMETERS SELECTION FOR EACH NOISE MECHANISM

Un pequeño parrafo que sustente porque se utilizaron los parametros de ruido dado (i.e p = 0.5, gamma = 0.5 etc) esto basicamente hace referencia al paper de Zhou.

VI. FINDING THE EXPERIMENTAL PRIVACY BUDGET

A. CENTRAL SETTING

La explicacion del experimento central, es basicamente puesto en texto la presentacion de semana 8

B. LOCAL SETTING

La explicacion del experimento loca, es basicamente puesto en texto la presentacion de semana 10

VII. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

La explicación de la medicion de las variables categoricas como forma de mostrar el ruido en acción. Es el resto de la presentación de semana 10 puesto en texto

VIII. EXPERIMENTAL RESULTS

A. FINDING THE EXPERIMENTAL PRIVACY BUDGET

1) Central Setting

Resultados del experimento central

2) Local Setting

Resultados del experimento local

B. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

Resultados de las mediciones con las distribuciones de probabilidad

IX. DISCUSSION AND CONCLUSION

En esta seccion voy a hablar de limitaciones encontradas dentro de los experimentos.

Entre ellas:

La dualidad entre C y L para la privacidad. (pros y contras) Que limitaciones tiene el experimento (analisis conjunto)

Tambien el analisis de los resultados de los mismos y que implicaciones tienen estos frente a la privacidad diferencial tanto clasica como cuantica.

ACKNOWLEDGMENTS

Esta seccion solo para la tesis o no se si pueda ir en el paper tambien

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